

Rotational Motion, solved

Step-by-step solutions to every practice problem in Chapter 7. Try each problem on your own before you read its solution.

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P7.1

(a) **Yes.** Torque changes angular velocity. It is not needed to *maintain* it. A wheel spinning on a frictionless axle keeps spinning at constant ω with zero torque, just as a puck on frictionless ice keeps moving with zero force. The Earth has rotated for billions of years with almost no torque acting on it.

(b) **Yes.** Two equal and opposite forces whose lines of action do not coincide form a **couple**. The net force is zero, so the CM does not accelerate, but the net torque is $\mathbf{F}\mathbf{d}$, where $\mathbf{d}$ is the separation of the lines of action. Example: your two hands turning a steering wheel, one pushing up and one pulling down.

P7.2

The **solid cylinder** reaches the bottom first. With $a = g \sin \theta / (1 + \beta)$:

- Solid cylinder, $\beta = \frac{1}{2}$: $a = \frac{2}{3}g \sin \theta$.
- Thin-walled pipe, $\beta = 1$: $a = \frac{1}{2}g \sin \theta$.

Mass does not matter. The mass cancels from $a = g \sin \theta / (1 + I/MR^2)$, so a hollow cylinder twice as heavy still loses. Only the *shape* matters: how far the mass is from the axis.

P7.3

The ice exerts negligible torque about the vertical axis, so the skater's angular momentum $\mathbf{L} = I\omega$ is conserved. Pulling the arms in moves mass closer to the axis, which reduces I , so ω must increase.

Kinetic energy is not conserved. $K = L^2/2I$ increases as I decreases. The extra energy comes from the skater's **muscles**. In her rotating frame, her hands feel an outward centrifugal pull, and drawing them inward against it does work. In the ground frame, she must supply the extra centripetal force needed as her hands speed up along a smaller circle.

P7.4

To open a door, you need a certain torque, $\tau = rF$ (pushing perpendicular to the door). For a given torque, the force needed is inversely proportional to the distance from the hinge:

$$\frac{F_{10 \text{ cm}}}{F_{80 \text{ cm}}} = \frac{80}{10} = 8.$$

Pushing 10 cm from the hinges takes **eight times** the force. That is why handles are put as far from the hinges as possible.

P7.5

(a) **Kinetic energy.** For a uniform disc:

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(50)(0.40)^2 = 4.0 \text{ kg m}^2, \quad \omega = 3000 \times \frac{2\pi}{60} = 314 \text{ rad/s}.$$

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}(4.0)(314)^2 = 1.97 \times 10^5 \text{ J} \approx 197 \text{ kJ}.$$

(b) **Braking torque.** The angular deceleration is $\alpha = 314/20 = 15.7 \text{ rad/s}^2$, so

$$\tau = I\alpha = 4.0(15.7) = 62.8 \text{ N m}.$$

(c) **Revolutions.** With uniform deceleration:

$$\theta = \bar{\omega}t = \frac{1}{2}(314)(20) = 3140 \text{ rad} = 500 \text{ rev}.$$

Check: the work done by the brake is $\tau\theta = 62.8(3140) = 1.97 \times 10^5 \text{ J} = K$. ✓

P7.6

(a) **Perpendicular axis through the centre.** Each mass is half a diagonal from the centre:

$r = 0.20\sqrt{2} \text{ m}$, so $r^2 = 0.080 \text{ m}^2$.

$$I = 4(0.50)(0.080) = 0.16 \text{ kg m}^2.$$

(b) **Along one side.** Two masses lie on the axis ($r = 0$). The other two are 0.40 m away:

$$I = 2(0.50)(0.40)^2 = 0.16 \text{ kg m}^2.$$

(c) **Along a diagonal.** Two masses lie on the axis. The other two are half a diagonal away, $r^2 = 0.080 \text{ m}^2$:

$$I = 2(0.50)(0.080) = 0.080 \text{ kg m}^2.$$

Check with the perpendicular-axis theorem. The two diagonals are perpendicular axes in the plane of the square, so $I_z = I_{d1} + I_{d2} = 0.080 + 0.080 = 0.16$. ✓

P7.7

Equations. For the bucket (down positive): $mg - T = ma$. For the windlass:

$$TR = I\alpha = \frac{1}{2}MR^2(a/R), \text{ so } T = \frac{1}{2}Ma.$$

(a) Combining:

$$a = \frac{mg}{m + \frac{1}{2}M} = \frac{5.0(9.8)}{5.0 + 5.0} = 4.9 \text{ m/s}^2.$$

$$(b) T = \frac{1}{2}Ma = \frac{1}{2}(10)(4.9) = 24.5 \text{ N}.$$

$$(c) v = \sqrt{2as} = \sqrt{2(4.9)(3.0)} = 5.4 \text{ m/s}.$$

Check by energy: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}(m + \frac{1}{2}M)v^2$, so $v^2 = 2(49)(3.0)/10 = 29.4$, giving $v = 5.4 \text{ m/s}$. ✓

P7.8

Rolling solid ball ($\beta = \frac{2}{5}$):

$$Mgh = \frac{1}{2}Mv^2 \left(1 + \frac{2}{5}\right) \Rightarrow v = \sqrt{\frac{10gh}{7}} = \sqrt{\frac{10(9.8)(1.5)}{7}} = \sqrt{21.0} = 4.58 \text{ m/s}.$$

Sliding block: $v = \sqrt{2gh} = \sqrt{29.4} = 5.42 \text{ m/s}$.

Why the difference. The rolling ball converts the same potential energy into *two* kinds of kinetic energy. Two-sevenths of it goes into rotation, $\frac{1}{2}I\omega^2$, and only five-sevenths into forward motion. Static friction makes the ball spin but does no net work, since the contact point does not slide. The block has no rotation, so all of its energy goes into forward motion.

P7.9

Balance. Take torques about the pivot. The plank's weight acts at the pivot, so it has no torque:

$$30(9.8)(1.5) = 40(9.8)x \Rightarrow x = \frac{30(1.5)}{40} = 1.125 \text{ m}$$

on the other side.

Pivot force. For vertical equilibrium, the pivot supports everything:

$$N = (30 + 40 + 10)(9.8) = 784 \text{ N}.$$

P7.10

Torques about the shoulder joint. This eliminates the unknown joint force. Only the perpendicular component of the deltoid force, $T \sin 15^\circ$, has a lever arm about the joint along the arm:

$$T \sin 15^\circ (0.15) = (4.0)(9.8)(0.30) = 11.76 \text{ N m}.$$

$$T = \frac{11.76}{0.15(0.2588)} = 303 \text{ N}.$$

Joint force. The deltoid pulls up and *towards* the body, so the joint must push the arm outward and, as it turns out, down.

- Horizontal: $J_x = T \cos 15^\circ = 303(0.966) = 293 \text{ N}$, away from the body.
- Vertical: $T \sin 15^\circ - mg + J_y = 0$, so $J_y = 39.2 - 78.4 = -39.2 \text{ N}$ (down).

$$|J| = \sqrt{293^2 + 39.2^2} = 295 \text{ N}.$$

Evaluate. Holding a 4 kg arm out needs a muscle force of about 300 N, nearly eight times the arm's weight, because the deltoid pulls almost along the arm. This is why holding your arms out is so tiring, and why shoulder joints are prone to wear.

P7.11

Forces. The floor provides N_f (up) and friction f (towards the wall). The frictionless wall provides only a horizontal force N_w . The weights are 196 N (ladder, at its midpoint) and 686 N (painter, at distance d along the ladder).

Force balance:

$$N_f = (20 + 70)(9.8) = 882 \text{ N}, \quad f = N_w.$$

The ladder is about to slip when $f = \mu_s N_f = 0.40(882) = 353 \text{ N}$, so $N_w = 353 \text{ N}$.

Torques about the base. The lever arm of N_w is $L \sin 60^\circ$, and the lever arm of each weight is (its distance along the ladder) $\times \cos 60^\circ$:

$$N_w(5.0) \sin 60^\circ = 196(2.5) \cos 60^\circ + 686 d \cos 60^\circ.$$

$$353(4.33) = 245 + 343 d \Rightarrow 1528 = 245 + 343 d \Rightarrow d = 3.74 \text{ m}.$$

The painter can climb 3.7 m along the ladder, about three-quarters of the way up, before it slips.

P7.12

(a) **Angular momentum.** There is no external torque about the axle.

- Initially: $I_i = \frac{1}{2}(200)(2.0)^2 + 50(2.0)^2 = 400 + 200 = 600 \text{ kg m}^2$.
- Finally, with the child at the centre: $I_f = 400 \text{ kg m}^2$.

$$\omega_f = \frac{I_i}{I_f} \omega_i = \frac{600}{400} \cdot \frac{2\pi}{4.0} = 1.5(1.571) = 2.36 \text{ rad/s}, \quad T_f = \frac{4.0}{1.5} = 2.67 \text{ s}.$$

(b) **Kinetic energy:**

$$K_i = \frac{1}{2}(600)(1.571)^2 = 740 \text{ J}, \quad K_f = \frac{1}{2}(400)(2.36)^2 = 1110 \text{ J}.$$

The kinetic energy increases by 370 J. The energy comes from the child's **muscles**. To walk inward, the child must pull against the merry-go-round's tendency to fling them outward, which in the rotating frame is the centrifugal force, and that takes work.

P7.13

Slice the sphere. Use a thin disc at height z above the centre, with thickness dz . Its radius is $r = \sqrt{R^2 - z^2}$, and its mass is

$$dm = \rho \pi r^2 dz = \rho \pi (R^2 - z^2) dz.$$

Each disc contributes $dI = \frac{1}{2} dm r^2 = \frac{1}{2} \rho \pi (R^2 - z^2)^2 dz$.

Integrate from $-R$ to R :

$$I = \frac{1}{2}\rho\pi \int_{-R}^R (R^4 - 2R^2z^2 + z^4) dz = \frac{1}{2}\rho\pi \left(2R^5 - \frac{4}{3}R^5 + \frac{2}{5}R^5\right) = \frac{1}{2}\rho\pi R^5 \cdot \frac{16}{15} = \frac{8\rho\pi R^5}{15}.$$

Express in terms of M . Substitute $\rho = M/(\frac{4}{3}\pi R^3) = 3M/(4\pi R^3)$:

$$I = \frac{8\pi R^5}{15} \cdot \frac{3M}{4\pi R^3} = \frac{24}{60}MR^2 = \frac{2}{5}MR^2. \checkmark$$

P7.14

While sliding, kinetic friction $\mu_k Mg$ acts backward at the contact point.

- Linear: $a = -\mu_k g$, so $v = v_0 - \mu_k g t$.
- Rotational: the friction torque about the CM is $\mu_k MgR = \frac{2}{5}MR^2\alpha$, so $\alpha = \frac{5\mu_k g}{2R}$ and $\omega = \frac{5\mu_k g}{2R}t$.

The friction slows the forward motion while spinning the ball up.

(a) Rolling begins when $v = \omega R$:

$$v_0 - \mu_k g t = \frac{5}{2}\mu_k g t \Rightarrow t = \frac{2v_0}{7\mu_k g}. \checkmark$$

(b) Rolling speed:

$$v = v_0 - \mu_k g \cdot \frac{2v_0}{7\mu_k g} = \frac{5}{7}v_0. \checkmark$$

A neat check: angular momentum about the contact point on the lane is conserved, because friction acts *at* that point. So $Mv_0R = Mv_fR + \frac{2}{5}MR^2(v_f/R)$, giving $v_f = \frac{5}{7}v_0$. $\checkmark$

(c) Sliding distance:

$$d = v_0 t - \frac{1}{2}\mu_k g t^2 = \frac{12v_0^2}{49\mu_k g}.$$

Numbers: $\mu_k g = 1.96 \text{ m/s}^2$.

$$t = \frac{2(8.0)}{7(1.96)} = 1.17 \text{ s}, \quad v = \frac{5}{7}(8.0) = 5.7 \text{ m/s}, \quad d = \frac{12(64)}{49(1.96)} = 8.0 \text{ m}.$$

(d) **Energy lost.**

$$K_f = \frac{1}{2} M v^2 \left(1 + \frac{2}{5}\right) = 0.7 M \left(\frac{5}{7} v_0\right)^2 = \frac{5}{14} M v_0^2.$$

As a fraction of $K_0 = \frac{1}{2} M v_0^2$, this is $\frac{5}{7}$. So $\frac{2}{7} \approx 29\%$ of the kinetic energy is lost to friction while the ball slides.

P7.15

(a) **Energy.** The CM falls $L/2$. The rod rotates about the fixed hinge, with $I = \frac{1}{3} M L^2$:

$$Mg \frac{L}{2} = \frac{1}{2} \cdot \frac{1}{3} M L^2 \omega^2 \Rightarrow \omega = \sqrt{\frac{3g}{L}}.$$

(b) **Speed of the free end:** $v_{\text{tip}} = \omega L = \sqrt{3gL}$. An object dropped from height L lands at $\sqrt{2gL}$, so the tip is faster by a factor $\sqrt{3/2} = 1.22$.

(c) **Acceleration when horizontal.** The torque of gravity about the hinge is $Mg(L/2)$:

$$\alpha = \frac{MgL/2}{ML^2/3} = \frac{3g}{2L}.$$

The tangential acceleration of the tip, which is vertically downward at this moment, is $a_{\text{tip}} = \alpha L = \frac{3}{2}g$, faster than free fall.

Chimneys. Every point beyond $\frac{2}{3}L$ from the base has a tangential acceleration greater than g . For the upper part of a falling chimney to keep up with rigid rotation, the lower part must *pull it down* faster than gravity alone would. That requires large bending stresses. Masonry is very weak in tension, so the chimney breaks, typically about a third to half of the way up, and the top part then lags behind.

P7.16

Equations. For the CM (down positive): $Mg - T = Ma$. For rotation about the CM: $TR = \frac{1}{2} MR^2 \alpha$. The string unwinds without slipping, so $a = R\alpha$. The rotation equation then gives $T = \frac{1}{2} Ma$.

Solve.

$$Mg - \frac{1}{2} Ma = Ma \Rightarrow a = \frac{2}{3}g = 6.5 \text{ m/s}^2, \quad T = \frac{1}{3} Mg.$$

Evaluate. The yo-yo falls at two-thirds of g , and the string carries one-third of its weight. A yo-yo with a small inner axle around which the string is wound, rather than the full radius, falls much more slowly. That design gives the long, slow spin that makes yo-yo tricks possible.

P7.17

Geometry. The spine is at 30° above horizontal. A vertical force applied at distance s along the spine has a lever arm $s \cos 30^\circ$ about the base. The muscle pulls at 12° to the spine, so its perpendicular component is $T \sin 12^\circ$, applied at $\frac{2}{3}L$.

(a) **Torques about L5/S1:**

$$T \sin 12^\circ \left(\frac{2}{3}L\right) = 450 \left(\frac{L}{2}\right) \cos 30^\circ + 200 L \cos 30^\circ.$$

L cancels:

$$T = \frac{\cos 30^\circ (225 + 200)}{\frac{2}{3} \sin 12^\circ} = \frac{0.866(425)}{0.667(0.208)} = \frac{368}{0.139} = 2.66 \times 10^3 \text{ N}.$$

(b) **Compression along the spine.** Resolve all the forces on the spine along its length. The joint reaction must balance them.

- The muscle's component along the spine, towards the base: $T \cos 12^\circ = 2660(0.978) = 2600 \text{ N}$.
- The weights' components along the spine, towards the base: $(450 + 200) \sin 30^\circ = 325 \text{ N}$.

$$C \approx 2600 + 325 = 2.9 \times 10^3 \text{ N}.$$

(The perpendicular components nearly cancel, leaving only a small shear force.)

Comparison. Body weight is $75(9.8) = 735 \text{ N}$, so the disc compression is about **4 times body weight**. That is below, but close to, the 3.4 kN guideline. A heavier load, a deeper bend, or holding the load farther from the body would exceed it. Squatting with an upright back cuts the lever arms by perhaps half, and with them the spinal load.

P7.18

(a) **Angular momentum about the pivot** is conserved. During the brief impact, the pivot exerts a large force on the rod, so linear momentum is **not** conserved. But that force acts *at* the pivot, so it has no torque about it. Gravity's torque is negligible during the very short impact. Kinetic energy is **not** conserved, because the collision is perfectly inelastic: the bullet embeds itself.

(b) **Angular momentum:**

$$mvL = \left(\frac{1}{3}ML^2 + mL^2\right)\omega.$$

$$0.010(300)(1.0) = (0.667 + 0.010)\omega \Rightarrow \omega = \frac{3.0}{0.677} = 4.43 \text{ rad/s}.$$

(c) **Energy after the impact.** Mechanical energy is conserved during the swing:

$$\frac{1}{2}I\omega^2 = \frac{1}{2}(0.677)(4.43)^2 = 6.65 \text{ J}.$$

When the rod swings through angle θ , its CM (at $L/2$) and the bullet (at L) rise:

$$\left(Mg\frac{L}{2} + mgL\right)(1 - \cos\theta) = (9.80 + 0.098)(1 - \cos\theta) = 9.90(1 - \cos\theta).$$

$$1 - \cos\theta = \frac{6.65}{9.90} = 0.672 \Rightarrow \cos\theta = 0.328 \Rightarrow \theta = 71^\circ.$$

Evaluate. Only $6.65/450 = 1.5\%$ of the bullet's kinetic energy survives the impact. Compare the ballistic pendulum (Example 6.9), where linear momentum was conserved because the cords could not exert a horizontal impulse. For a rod on a rigid pivot, it is angular momentum that is conserved.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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